

RETAIL SALES ANALYTICS

Technical Project Report

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Tools Used:

Power BI | Excel

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1. Executive Summary

This report documents the end-to-end analytical process applied to a South African retail sales dataset. The project involved raw data ingestion, structured cleaning and transformation in SQL, exploratory and descriptive analysis, and interactive visualization in Microsoft Power BI.

The dataset comprises 76,226 transaction records spanning January 2022 to December 2024 across seven product categories, eight South African provinces, and five payment methods. Total confirmed revenue across the period amounts to R447,634,562.

Metric	Value	Notes
Total Transactions	76,226	Includes all statuses
Distinct Transactions	75,001	After deduplication
Total Revenue	R447,634,562	All order statuses
Average Order Value	R5,968	Per transaction
Average Rating	3.00 / 5	Uniform across categories
Date Range	Jan 2022 – Dec 2024	3 full years

2. Data Source & Pipeline Overview

2.1 Source Data

The source dataset is a single flat-file CSV representing retail transaction records for a South African multi-category retailer. The raw file was titled retail_sales_messy and contained deliberate data quality issues introduced for cleaning practice.

Attribute	Detail
File format	CSV (comma-separated values)
Raw row count	~76,000+ rows including duplicates and nulls
Columns (raw)	30 fields including transaction, customer, product, order, and rating data
Geographic scope	8 South African provinces
Time period	January 2022 to December 2024

2.2 Pipeline Architecture

The data pipeline followed a linear ETL (Extract, Transform, Load) approach:

- Raw CSV ingested into SQL environment for structured cleaning and transformation
- Cleaned dataset exported and loaded into Power BI Desktop for modelling and visualization
- Excel used for supplementary data exploration and formula-based validation

3. SQL Data Cleaning Process

3.1 Issues Identified in Raw Data

The following categories of data quality issues were identified and addressed during the SQL cleaning phase:

#	Issue Type	Affected Field(s)	Resolution
1	Duplicate records	transaction_id	Identified 1,225 duplicates; retained first occurrence
2	Mixed date formats	order_date	Standardized to YYYY-MM-DD using CASE/CONVERT logic
3	Null / missing values	region, payment_method, customer_name	Replaced with 'Unknown' label for traceability
4	Inconsistent categorization	payment_method (Debit vs Debit Card)	Partially resolved; 374 rows remain as 'Debit'
5	Revenue calculation mismatch	total_amount vs calculated_total	mismatch_flag column added to audit discrepancies
6	Unparsed date components	day, month, year, day_of_week	Extracted from order_date using date functions
7	Missing rating column name	rating.1 (renamed from duplicate)	Retained as-is; flagged for renaming

3.2 Derived / Calculated Columns

The following columns were derived or engineered during the SQL transformation phase:

- **revenue** = calculated_total_amount
- **Audit flag:** mismatch_flag
- **Binary flag:** returned_flag
- **Binary flag:** finalized_orders_flag
- **Calculated:** Age Group
- **Extracted:** Month Number / Day Number

Note on order_date data type

The order_date column remains stored as a String (text) type in the Power BI model. This is a known limitation – date-based DAX time intelligence functions (e.g. DATEADD, SAMEPERIODLASTYEAR) require a proper Date data type. A Date table with an explicit date column is recommended for future iterations.

4. Data Quality Findings

4.1 Completeness

After cleaning, the following residual data quality gaps remain documented:

Field	Issue	Record Count	% of Total
region	'Unknown' label	3,709	4.9%
payment_method	'Unknown' label	5,933	7.8%
payment_method	'Debit' (not 'Debit Card')	374	0.5%
order_date (year/month)	Null – unparseable dates	3,060	4.0%

4.2 Consistency

- Average customer rating is 3.00/5 across all seven product categories with virtually no variance. This uniformity is atypical of real-world rating data and is a known characteristic of this dataset.
- Revenue distribution across regions is broadly consistent, with Eastern Cape (R85.1M) and Gauteng (R84.2M) marginally ahead of other provinces.
- Monthly transaction volumes are stable at approximately 2,000 per month throughout the three-year period, suggesting the data was synthetically generated.

4.3 Data Quality Score

Based on the above findings, the dataset is estimated to be approximately 87% complete and consistent post-cleaning, with residual gaps documented and flagged rather than silently dropped.

5. Power BI Data Model

5.1 Model Structure

The Power BI semantic model is built on a single deformatized flat table loaded from the cleaned SQL output. The model does not implement a star's schema at this stage; all dimensions (customer, product, date, geography) are embedded within the single fact table.

Property	Value
Table name (internal)	retail_sales_messy 1(retail_sales_cleaned) (1)
Total columns	34 (30 source + 4 calculated)
Relationships	0 (single-table model)
Compatibility level	1600
Model state	Unprocessed (requires refresh)

5.2 Column Inventory

The following table documents all 34 columns in the model:

Column	Data Type	Calculated	Notes
transaction_id	String	No	Primary identifier
customer_id	String	No	
customer_name	String	No	May be 'Unknown'
age	Integer	No	
gender	String	No	
city	String	No	
region	String	No	May be 'Unknown'
category	String	No	7 categories
product_name	String	No	
quantity	Integer	No	
unit_price	Decimal	No	
discount_pct	Decimal	No	
total_amount	Decimal	No	Source total
calculated_total_amount	Decimal	No	Derived: qty × price × (1-disc)
revenue	Decimal	No	Canonical revenue field
average_order_value	Decimal	No	Redundant – also a measure
mismatch_flag	String	No	Audit: total vs calculated

payment_method	String	No	May be 'Unknown' or 'Debit'
order_date	String	No	⚠ Should be Date type
day	Integer	No	Extracted from order_date
month	Integer	No	Extracted from order_date
year	Integer	No	Extracted from order_date
day_of_week	String	No	
month_name	String	No	
order_status	String	No	Completed / Returned / Pending / Cancelled
returned_flag	Integer	No	Binary: 1=returned
finalized_orders_flag	Integer	No	Binary: 1=finalized
store_id	String	No	
sales_rep	String	No	
rating.1	Decimal	No	⚠ Should be renamed to rating
Month Number	Integer	Yes	Calculated column
Age Group	String	Yes	Calculated column – bins by age
Day Number	Integer	Yes	Calculated column
new_year	String	Yes	Calculated column – purpose unclear

6. DAX Measures Catalogue

The model contains 8 custom DAX measures. All measures currently reside in the single source table with no display folder organization.

Measure Name	Purpose	Result	Notes
Total Transactions	COUNT of transaction_id	76,226	Includes all statuses
Transactions Distinct	DISTINCTCOUNT of transaction_id	75,001	After duplicate removal
Revenue2	SUM of revenue column	R447,634,562	⚠ Name should be 'Total Revenue'
Average Order Value	Average transaction value	R5,968	Redundant column also exists
Average Spend	Like AOV	R5,860	Slightly different calculation
Average Rating	AVERAGE of rating.1	3.00 / 5	Uniform across all categories
Category Order Rank	RANKX by category revenue	Dynamic	Used in category visuals
Top 3 Categories	Filters to rank <= 3	Dynamic	Uses Category Order Rank

6.1 Recommended Additional Measures

- Total Revenue (rename Revenue2 for clarity)
- Return Rate % = DIVIDE([Returned Transactions], [Total Transactions])
- Completion Rate % = DIVIDE([Completed Transactions], [Total Transactions])
- YoY Revenue Growth = using SAMEPERIODLASTYEAR (requires Date table)

7. Analytical Findings

7.1 Revenue by Category

Category	Revenue	Transactions	Revenue Share
Electronics	R234,753,605	10,836	52.4%
Home & Garden	R104,819,720	10,788	23.4%
Sports	R51,263,845	10,980	11.5%
Clothing	R23,345,205	10,888	5.2%
Health & Beauty	R15,512,271	10,964	3.5%
Toys	R13,960,024	10,865	3.1%
Food & Beverage	R3,979,893	10,905	0.9%

Key Insight

Electronics dominates with 52.4% of total revenue despite similar transaction volumes across all categories (~10,800–11,000 each). This confirms that Electronics has a significantly higher average unit price rather than higher purchase frequency.

7.2 Revenue by Region

Region	Revenue	Transactions	Rank
Eastern Cape	R85,073,610	14,693	1st
Gauteng	R84,160,456	14,345	2nd
Northern Cape	R43,595,559	7,273	3rd
KwaZulu–Natal	R43,004,470	7,338	4th
Mpumalanga	R42,861,847	7,164	5th
Western Cape	R42,795,211	7,240	6th
Limpopo	R42,275,571	7,252	7th
Free State	R41,298,851	7,212	8th

7.3 Order Status Breakdown

Status	Transactions	Revenue	% of Total
Completed	57,163	R336,580,162	75.0%
Returned	7,655	R44,377,194	10.0%
Pending	7,564	R43,738,551	9.9%
Cancelled	3,844	R22,938,654	5.0%

8. Known Limitations & Next Steps

8.1 Current Limitations

- Single flat table model – no star schema; dimension tables for Date, Customer, Product, and Geography would significantly improve analytical flexibility
- order_date stored as String – prevents use of native Power BI date hierarchies and DAX time intelligence functions
- No date table – SAMEPERIODLASTYEAR, DATEADD, and similar functions are unavailable without an explicit Date dimension
- Residual 'Unknown' values in region (4.9%) and payment_method (7.8%) reduce segmentation accuracy
- rating.1 column name is non-descriptive – a legacy artefact from the data import process
- Revenue2 measure name does not clearly communicate its purpose
- No display folders on measures – model will become difficult to navigate as measure count grows

8.2 Recommended Next Steps

- Rename table to 'Sales' and clean up all column names (rating.1 → rating, Revenue2 → Total Revenue)
- Convert order_date to Date data type and create a dedicated Date dimension table
- Implement a star schema: separate Customers, Products, Geography, and Date tables
- Add Return Rate % and Completion Rate % DAX measures
- Organize measures into display folders (Revenue, Volume, Customer, Quality)
- Investigate and resolve the Debit vs Debit Card payment method inconsistency
- Review the purpose of the new_year calculated column – remove if redundant

9. Appendix

A. Tools & Technologies

Tool	Version / Platform	Role in Project
Microsoft Excel	Excel 365	Supplementary data validation and exploration
Power BI Desktop	Latest (June 2026)	Data modelling, DAX measures, visualization
Power BI MCP Developer	powerbi-modeling-mcp	Programmatic model inspection

B. Glossary

- AOV – Average Order Value: total revenue divided by total number of transactions
- DAX – Data Analysis Expressions: formula language used in Power BI for calculated columns and measures
- ETL – Extract, Transform, Load: the process of moving and transforming data from source to destination
- Star Schema – a data modelling pattern with a central fact table connected to multiple dimension tables
- TMDL – Tabular Model Definition Language: the serialization format used by Power BI for model definitions